Course 6013C | Semester VI | Theory | 4 Credits

Harbour, Tunnel & Airport Engineering

A complete syllabus-aligned handbook with clear explanations, practical or exam guidance, Malayalam-support notes, diagrams, activities and revision tools.

Programmes	**Civil Engineering**
Contact hours	**60**
Teaching scheme	**3:1:0:0**
Credits	**4**
CIE / ESE	**Theory CIE 40 / Theory ESE 60**
Self-learning	**5.5 h/week**

CURRICULUM INTEGRITY

Official Source Declaration and Verified Inconsistencies

The attached Revision 2026 index identifies Course 6013C as Harbour, Tunnel & Airport Engineering in Semester VI for Civil Engineering. The official SITTTR detailed course page was checked at the indexed link and returned “Requested file not found.” With the user's approved public-source approach, this handbook is transparently based on NPTEL IIT Madras’s Port and Harbour Structures course, NPTEL IIT Roorkee’s Transportation Engineering II course and the MSBTE polytechnic syllabus. It does not claim to reproduce official REV2026 detailed-syllabus wording.

Source treatment: Teaching scheme, credits and assessment values are provisional placeholders aligned to neighbouring handbook format. Harbour layouts, berthing structure designs, tunnel linings and airport runway orientations must be based on approved engineering designs, current maritime/tunnel/aviation codes, authoritative site data and competent professional review.

Verified source note: Verified sources support port planning, harbour layout, breakwaters, dredging, berthing structures, drydocks, airport site selection, runway orientation, visual aids and tunneling basics. The four-module sequence is a learning path, not an asserted official module split.

COURSE DASHBOARD

What You Are Expected to Learn

60
Contact hours

4
Credits

Theory CIE 40
CIE

Theory ESE 60
ESE

Course introduction

Harbour, Tunnel & Airport Engineering studies the specialized infrastructure required for maritime, underground and air transportation. It develops the ability to plan harbour layouts, design berthing structures, understand tunneling methods and ventilation, and coordinate airport site selection and runway design while respecting the technical and safety boundaries of transport engineering.

> **Malayalam support:** □ handbook □□□□□□□□□□□□ syllabus topic □□□□□ □□□□□□□□□□; □□□□□□□□□ principle, diagram/procedure, application, common mistake □□□□□□ □□□□□□□□□ □□□□□□□□□.

Prerequisite foundation

Transportation engineering, fluid mechanics, soil mechanics, basic surveying, engineering graphics and structural mechanics.

Use prerequisites only as a revision bridge. The current course outcomes and detailed syllabus define the required performance.

Teaching and assessment scheme

L	T	P	S	Self-learning/week	Contact hours	Credits	CIE	ESE
3	1	0	0	5.5 h/week	60	4	Theory CIE 40	Theory ESE 60

STUDY METHOD

How to Use This Handbook

1 · Understand

Read terms, conditions and purpose; make a short definition in your own words.

2 · Visualise

Follow the diagram, circuit, flow or procedure and label it accurately.

3 · Apply

Use the method block, calculation or practical checklist without skipping units or safety checks.

4 · Revise

Use questions, quick cards and common-mistake notes before examination.

OFFICIAL STATEMENTS

Objectives and Course Outcomes

Official course objectives

1. Explain harbour engineering fundamentals, classification, site selection and the components of ports and harbours.
2. Explain marine structures, including breakwaters, docks, quays and wharves, along with dredging and navigational aids.
3. Explain tunnel engineering principles, shapes, tunneling methods in soil and rock, and tunnel services.
4. Explain airport engineering planning, site selection, runway/taxiway design and visual aids for aircraft operation.

Student-friendly meaning

You should be able to recognise the relevant system or material, explain its principle, communicate through a correct diagram/table where needed and follow a defensible solution or practical method.

> **Malayalam support:** CO □□□□□□ exam-□ □□□□□ □□□□□□□□ □□□□□□□ □□□□□ □□□□□□□□□. Define, explain, apply □□□□□ action words □□□□□□□□□□□□.

Official Course Outcomes

CO	Official outcome	Bloom level
CO1	Explain harbour planning, classification and the functional components of ports and harbours.	Understand
CO2	Explain the design and function of breakwaters, docks, quays and other marine structures.	Understand
CO3	Explain tunneling methods, tunnel lining and the essential services like ventilation and lighting.	Understand
CO4	Explain airport site selection, runway orientation and the geometric design of airport facilities.	Understand

Official CO-PO articulation matrix

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7–PO11
CO1	3	2	2	2	2	–	–
CO2	3	2	2	2	2	–	–
CO3	3	2	2	2	2	–	–
CO4	3	2	2	2	2	–	–

SYLLABUS COVERAGE

Curriculum Coverage Map

All official major topics mapped to handbook chapters

Module	Title	Official major topics	Hours	CO	Handbook section
1	Harbour Engineering and Port Planning	Classification of harbours (natural, semi-natural, artificial); site selection for harbours; port planning and layout; ships and their characteristics; harbour components (entrance channel, turning circle).	Approx. 14 hours	CO1	Module 1
2	Marine Structures, Docks and Navigational Aids	Breakwaters (types and design); berthing structures (quays, wharves, piers, dolphins); docks (wet docks, dry docks, floating docks); dredging methods; navigational aids (lighthouses, buoys).	Approx. 16 hours	CO2	Module 2
3	Tunnel Engineering and Services	Introduction to tunneling; tunnel shapes and sizes; tunneling methods in soil (shield, pipe jacking) and rock (drill and blast); tunnel lining and drainage; ventilation and lighting.	Approx. 16 hours	CO3	Module 3
4	Airport Engineering and Visual Aids	Airport planning and site selection; aircraft characteristics; runway orientation (wind rose); runway and taxiway geometric design; airport obstructions; visual aids (markings, lighting).	Approx. 14 hours	CO4	Module 4

LEARNING ROADMAP

How the Modules Connect

Harbour Engineering and Port Planning

Classification of harbours (natural, semi-natural, artificial); site selection for harbours; port planning and layout; ships and their characteristics; harbour components (entrance channel, turning circle).

CO1 · Approx. 14 hours

Marine Structures, Docks and Navigational Aids

Breakwaters (types and design); berthing structures (quays, wharves, piers, dolphins); docks (wet docks, dry docks, floating docks); dredging methods; navigational aids (lighthouses, buoys).

CO2 · Approx. 16 hours

Tunnel Engineering and Services

Introduction to tunneling; tunnel shapes and sizes; tunneling methods in soil (shield, pipe jacking) and rock (drill and blast); tunnel lining and drainage; ventilation and lighting.

CO3 · Approx. 16 hours

Airport Engineering and Visual Aids

Airport planning and site selection; aircraft characteristics; runway orientation (wind rose); runway and taxiway geometric design; airport obstructions; visual aids (markings, lighting).

CO4 · Approx. 14 hours

MODULE 1

Harbour Engineering and Port Planning

Classification of harbours (natural, semi-natural, artificial); site selection for harbours; port planning and layout; ships and their characteristics; harbour components (entrance channel, turning circle).

Approx. 14 hours **CO1** **Understand · Apply**

Learning goals

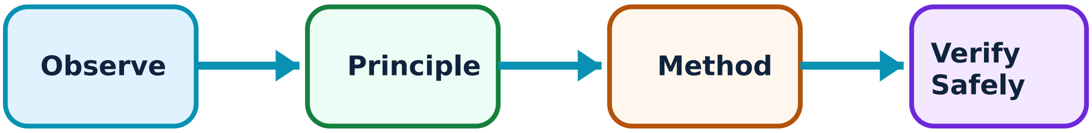
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Harbour Engineering and Port Planning: identify the system, state its principle, follow the correct method and verify the result safely.

1. Harbour classification and site selection−

Harbours provide shelter for ships. Natural harbours are formed by land contours, while artificial ones require breakwaters. Site selection considers water depth, protection from waves, soil conditions, land availability and connectivity to the hinterland.

Study and application method

Classify three types of harbours based on their origin and list four factors to consider when selecting a site for a new commercial port.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Classify three types of harbours based on their origin and list four factors to consider when selecting a site for a new commercial port.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Harbour sites are sensitive to environmental and coastal changes. Do not propose sites without authorised bathymetric surveys and environmental impact studies.

2. Port planning and harbour layout−

Port planning organises the water and land areas for efficient ship handling and cargo transfer. The layout includes the entrance channel, turning basin for ship maneuvering, and berthing areas, designed according to the size and type of ships served.

Study and application method

Sketch a conceptual harbour layout showing the entrance channel, turning circle and berthing area; explain the role of a 'turning basin'.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

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Common mistake / precaution: Layouts must accommodate future ship size trends and safety requirements. Inadequate channel depth or turning space can lead to maritime accidents.

Module 1 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 2

Marine Structures, Docks and Navigational Aids

Breakwaters (types and design); berthing structures (quays, wharves, piers, dolphins); docks (wet docks, dry docks, floating docks); dredging methods; navigational aids (lighthouses, buoys).

Approx. 16 hours | **CO2** | **Understand · Apply**

Learning goals

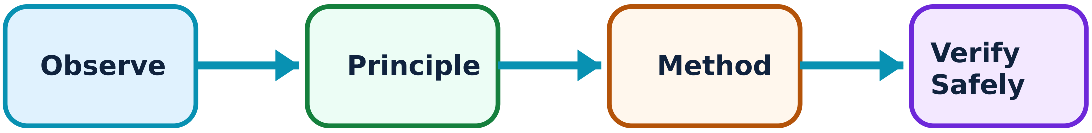
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Marine Structures, Docks and Navigational Aids: identify the system, state its principle, follow the correct method and verify the result safely.

1. Breakwaters and berthing structures−

Breakwaters protect the harbour from wave action. Berthing structures like quays and wharves provide platforms for ship loading/unloading. Dolphins are isolated structures for mooring or protecting other structures from ship impact.

Study and application method

Compare the functions of a quay wall and a wharf; describe the working principle of a mound-type breakwater.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Compare the functions of a quay wall and a wharf; describe the working principle of a mound-type breakwater.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Marine structures face extreme wave and seismic forces. Design and construction must follow authorised maritime engineering codes and use corrosion-resistant materials.

2. Docks, dredging and navigational aids−

Docks maintain water levels for ship handling (wet docks) or provide dry areas for repair (dry docks). Dredging maintains required water depths. Navigational aids like lighthouses and buoys guide ships safely into the harbour.

Study and application method

Describe the difference between a dry dock and a floating dock; list three types of navigational aids used in harbour entrance channels.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Describe the difference between a dry dock and a floating dock; list three types of navigational aids used in harbour entrance channels.

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Common mistake / precaution: Dredging can impact marine ecosystems. Navigational aids must strictly comply with international maritime standards (e.g., IALA) for safe passage.

Module 2 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 3

Tunnel Engineering and Services

Introduction to tunneling; tunnel shapes and sizes; tunneling methods in soil (shield, pipe jacking) and rock (drill and blast); tunnel lining and drainage; ventilation and lighting.

Approx. 16 hours | **CO3** | **Understand · Apply**

Learning goals

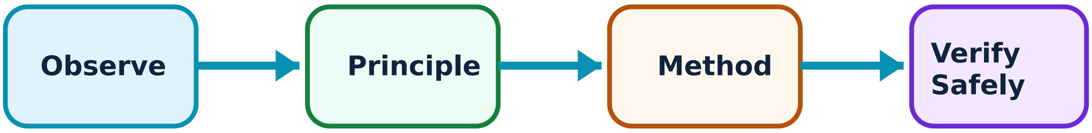
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Tunnel Engineering and Services: identify the system, state its principle, follow the correct method and verify the result safely.

1. Tunneling methods and lining−

Tunneling methods depend on the ground conditions. Rock tunneling often uses drill-and-blast or Tunnel Boring Machines (TBM). Soil tunneling uses shields or pipe jacking to prevent collapse. Linings (concrete, steel) provide structural support and prevent water ingress.

Study and application method

Compare the 'drill and blast' method with the 'shield' method of tunneling; explain why tunnel lining is necessary in soft soil.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Compare the 'drill and blast' method with the 'shield' method of tunneling; explain why tunnel lining is necessary in soft soil.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Tunneling is high-risk engineering. Ground stability and water pressure must be continuously monitored by authorised geotechnicians to prevent collapses.

2. Tunnel services and safety−

Safe tunnel operation requires continuous ventilation (to remove exhaust gases), adequate lighting, effective drainage and fire-safety systems. Emergency exits and communication systems are critical for user safety in long tunnels.

Study and application method

Describe the two main types of tunnel ventilation systems; list the essential safety features required for a road tunnel longer than 500m.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Describe the two main types of tunnel ventilation systems; list the essential safety features required for a road tunnel longer than 500m.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Tunnel fires and ventilation failures are catastrophic. Do not bypass safety systems, and ensure all services are maintained by authorised personnel.

Module 3 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 4

Airport Engineering and Visual Aids

Airport planning and site selection; aircraft characteristics; runway orientation (wind rose); runway and taxiway geometric design; airport obstructions; visual aids (markings, lighting).

Approx. 14 hours | CO4 | Understand · Apply

Learning goals

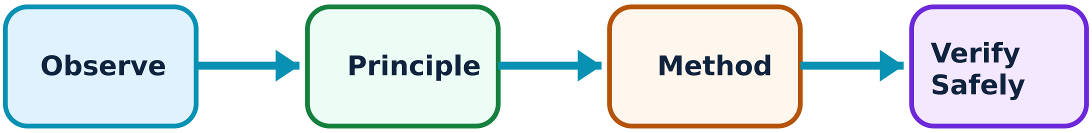
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Airport Engineering and Visual Aids: identify the system, state its principle, follow the correct method and verify the result safely.

1. Airport planning and runway design−

Airport planning balances traffic needs with site constraints. Runway orientation is determined by the prevailing wind direction using a wind rose diagram to maximize safety. Geometric design covers runway length, width, gradients and taxiway layouts.

Study and application method

Explain how a 'wind rose' diagram is used to determine the best orientation for a runway; list the factors that affect the required runway length.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Explain how a 'wind rose' diagram is used to determine the best orientation for a runway; list the factors that affect the required runway length.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Airport sites require clear approach paths. Do not plan airports near high-rise structures or in areas with poor visibility or extreme weather without authorised studies.

2. Visual aids and airport safety−

Visual aids include runway markings (threshold, centre line) and lighting (approach, runway edge, PAPI) to guide pilots during landing and takeoff. Safety also involves managing obstructions in the 'imaginary surfaces' around the airport.

Study and application method

Sketch the standard markings for a commercial runway threshold; explain the purpose of PAPI (Precision Approach Path Indicator) lights.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Sketch the standard markings for a commercial runway threshold; explain the purpose of PAPI (Precision Approach Path Indicator) lights.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Aviation visual aids must strictly comply with national and international civil aviation regulations (e.g., DGCA/ICAO). Incorrect lighting or markings can lead to aviation disasters.

Module 4 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

FORMULA AND PROCEDURE BANK

Rapid Recall Bank

Recall 1

State symbols and SI units before substitution.

Recall 2

Draw the circuit, system boundary, vector or process flow before reasoning.

Recall 3

For laboratory work: aim → apparatus → diagram → procedure → observation → calculation → result → precautions.

Recall 4

For a graph: label axes, units, scale, operating region and conclusion.

Recall 5

For a comparison: use construction, working, result, advantage and limitation.

Recall 6

For an extended answer: definition/principle, labelled diagram, working steps, application and a caution/limitation.

DIAGRAM LIBRARY

Diagram Library for Rapid Revision

Module 1: Harbour Engineering and Port Planning

Use the concept flow to revise observation, principle, method and verification.

Module 2: Marine Structures, Docks and Navigational Aids

Use the concept flow to revise observation, principle, method and verification.

Module 3: Tunnel Engineering and Services

Use the concept flow to revise observation, principle, method and verification.

Module 4: Airport Engineering and Visual Aids

Use the concept flow to revise observation, principle, method and verification.

SELF-LEARNING

Official Self-Learning and Student Activities

Structured activity

Classify harbours based on their location and purpose in a conceptual table.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Sketch the cross-section of a mound breakwater and a quay wall.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Compare two tunneling methods based on their suitability for different soil conditions.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Draw a wind rose diagram conceptually and identify the optimum runway orientation.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

List the visual aids and safety equipment required for a domestic airport terminal and runway.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Activity discipline: The supplied PDF assigns the listed self-learning hours. This handbook preserves the activity direction; the teacher's approved schedule and safety instructions govern execution.

EXAMINATION READINESS

Assessment Methodology and Examination Pattern

Official assessment structure		
Component	**Official mark / conversion**	**Student evidence**
Attendance	5 marks	End-of-semester attendance component.
Self-learning activities	15 marks	Module-linked activities.
Series examinations	20 + 20 converted	Two 50-mark series examinations.
CIE total	Theory CIE 40	Continuous internal evaluation.
Written ESE	Theory ESE 60	2.5 hours; official Part A / Part B / Part C pattern.

Series-test pattern

The supplied theory pattern uses Part A (1-mark), Part B (3-mark with choice) and Part C (7-mark) distribution across the stated modules. Practical courses follow the supplied practical-test scheme.

Examination evidence

Show a complete method, correct units/conditions, clear diagrams or tables, a result/conclusion and safety awareness for any apparatus or process involved.

EXAM PRACTICE

Module-wise Questions with Answers

Module 1 — Harbour Engineering and Port Planning

Explain Harbour classification and site selection and state one correct method, application or precaution.−

Harbours provide shelter for ships. Natural harbours are formed by land contours, while artificial ones require breakwaters. Site selection considers water depth, protection from waves, soil conditions, land availability and connectivity to the hinterland.

Answer framework: Classify three types of harbours based on their origin and list four factors to consider when selecting a site for a new commercial port.

Important point: Harbour sites are sensitive to environmental and coastal changes. Do not propose sites without authorised bathymetric surveys and environmental impact studies.

Explain Port planning and harbour layout and state one correct method, application or precaution.−

Port planning organises the water and land areas for efficient ship handling and cargo transfer. The layout includes the entrance channel, turning basin for ship maneuvering, and berthing areas, designed according to the size and type of ships served.

Answer framework: Sketch a conceptual harbour layout showing the entrance channel, turning circle and berthing area; explain the role of a 'turning basin'.

Important point: Layouts must accommodate future ship size trends and safety requirements. Inadequate channel depth or turning space can lead to maritime accidents.

Module 2 — Marine Structures, Docks and Navigational Aids

Explain Breakwaters and berthing structures and state one correct method, application or precaution.−

Breakwaters protect the harbour from wave action. Berthing structures like quays and wharves provide platforms for ship loading/unloading. Dolphins are isolated structures for mooring or protecting other structures from ship impact.

Answer framework: Compare the functions of a quay wall and a wharf; describe the working principle of a mound-type breakwater.

Important point: Marine structures face extreme wave and seismic forces. Design and construction must follow authorised maritime engineering codes and use corrosion-resistant materials.

Explain Docks, dredging and navigational aids and state one correct method, application or precaution.−

Docks maintain water levels for ship handling (wet docks) or provide dry areas for repair (dry docks). Dredging maintains required water depths. Navigational aids like lighthouses and buoys guide ships safely into the harbour.

Answer framework: Describe the difference between a dry dock and a floating dock; list three types of navigational aids used in harbour entrance channels.

Important point: Dredging can impact marine ecosystems. Navigational aids must strictly comply with international maritime standards (e.g., IALA) for safe passage.

Module 3 — Tunnel Engineering and Services

Explain Tunneling methods and lining and state one correct method, application or precaution.−

Tunneling methods depend on the ground conditions. Rock tunneling often uses drill-and-blast or Tunnel Boring Machines (TBM). Soil tunneling uses shields or pipe jacking to prevent collapse. Linings (concrete, steel) provide structural support and prevent water ingress.

Answer framework: Compare the 'drill and blast' method with the 'shield' method of tunneling; explain why tunnel lining is necessary in soft soil.

Important point: Tunneling is high-risk engineering. Ground stability and water pressure must be continuously monitored by authorised geotechnicians to prevent collapses.

Explain Tunnel services and safety and state one correct method, application or precaution.−

Safe tunnel operation requires continuous ventilation (to remove exhaust gases), adequate lighting, effective drainage and fire-safety systems. Emergency exits and communication systems are critical for user safety in long tunnels.

Answer framework: Describe the two main types of tunnel ventilation systems; list the essential safety features required for a road tunnel longer than 500m.

Important point: Tunnel fires and ventilation failures are catastrophic. Do not bypass safety systems, and ensure all services are maintained by authorised personnel.

Module 4 — Airport Engineering and Visual Aids

Explain Airport planning and runway design and state one correct method, application or precaution.−

Airport planning balances traffic needs with site constraints. Runway orientation is determined by the prevailing wind direction using a wind rose diagram to maximize safety. Geometric design covers runway length, width, gradients and taxiway layouts.

Answer framework: Explain how a 'wind rose' diagram is used to determine the best orientation for a runway; list the factors that affect the required runway length.

Important point: Airport sites require clear approach paths. Do not plan airports near high-rise structures or in areas with poor visibility or extreme weather without authorised studies.

Explain Visual aids and airport safety and state one correct method, application or

precaution.−

Visual aids include runway markings (threshold, centre line) and lighting (approach, runway edge, PAPI) to guide pilots during landing and takeoff. Safety also involves managing obstructions in the 'imaginary surfaces' around the airport.

Answer framework: Sketch the standard markings for a commercial runway threshold; explain the purpose of PAPI (Precision Approach Path Indicator) lights.

Important point: Aviation visual aids must strictly comply with national and international civil aviation regulations (e.g., DGCA/ICAO). Incorrect lighting or markings can lead to aviation disasters.

PRACTICE ONLY

Practice Model Paper

Important: This practice paper is based on the official pattern in the supplied syllabus. It is **not** an official SITTTR model question paper.

Part A — short response

1. State one key definition from Module 1: Harbour Engineering and Port Planning.
2. Identify one diagram, observation, formula or safety condition relevant to Module 1.
3. State one key definition from Module 2: Marine Structures, Docks and Navigational Aids.
4. Identify one diagram, observation, formula or safety condition relevant to Module 2.
5. State one key definition from Module 3: Tunnel Engineering and Services.
6. Identify one diagram, observation, formula or safety condition relevant to Module 3.
7. State one key definition from Module 4: Airport Engineering and Visual Aids.
8. Identify one diagram, observation, formula or safety condition relevant to Module 4.

Part B — structured explanation

1. Explain a selected procedure/principle from Module 1 with a labelled diagram, table or method.
2. Explain a selected procedure/principle from Module 2 with a labelled diagram, table or method.
3. Explain a selected procedure/principle from Module 3 with a labelled diagram, table or method.
4. Explain a selected procedure/principle from Module 4 with a labelled diagram, table or method.

Part C — extended response / practical task

1. Develop a complete answer or practical plan for: Classification of harbours (natural, semi-natural, artificial); site selection for harbours; port planning and layout; ships and their characteristics; harbour components (entrance channel, turning circle)..

2. Develop a complete answer or practical plan for: Breakwaters (types and design); berthing structures (quays, wharves, piers, dolphins); docks (wet docks, dry docks, floating docks); dredging methods; navigational aids (lighthouses, buoys)..
3. Develop a complete answer or practical plan for: Introduction to tunneling; tunnel shapes and sizes; tunneling methods in soil (shield, pipe jacking) and rock (drill and blast); tunnel lining and drainage; ventilation and lighting..
4. Develop a complete answer or practical plan for: Airport planning and site selection; aircraft characteristics; runway orientation (wind rose); runway and taxiway geometric design; airport obstructions; visual aids (markings, lighting)..

LAST-MILE REVISION

Quick Revision

Module 1: Harbour Engineering and Port Planning

Classification of harbours (natural, semi-natural, artificial); site selection for harbours; port planning and layout; ships and their characteristics; harbour components (entrance channel, turning circle).

- Match each topic to CO1.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Module 2: Marine Structures, Docks and Navigational Aids

Breakwaters (types and design); berthing structures (quays, wharves, piers, dolphins); docks (wet docks, dry docks, floating docks); dredging methods; navigational aids (lighthouses, buoys).

- Match each topic to CO2.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Module 3: Tunnel Engineering and Services

Introduction to tunneling; tunnel shapes and sizes; tunneling methods in soil (shield, pipe jacking) and rock (drill and blast); tunnel lining and drainage; ventilation and lighting.

- Match each topic to CO3.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Module 4: Airport Engineering and Visual Aids

Airport planning and site selection; aircraft characteristics; runway orientation (wind rose); runway and taxiway geometric design; airport obstructions; visual aids (markings, lighting).

- Match each topic to CO4.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Common mistakes: Writing an unlabelled diagram; ignoring units, polarity or conditions; treating a practical result as valid without observations; confusing a definition with an application; and omitting a safety precaution where the topic uses apparatus, current, heat, chemicals or tools.

Exam checklist: Read the command word; answer every part; draw clean labelled diagrams; use a clear calculation/procedure; state result with units or observation; and reserve two minutes for checking.

VOCABULARY

Glossary

Important terms from the syllabus	
Term / topic	**One-line meaning**
Harbour classification and site selection	Harbours provide shelter for ships.
Port planning and harbour layout	Port planning organises the water and land areas for efficient ship handling and cargo transfer.
Breakwaters and berthing structures	Breakwaters protect the harbour from wave action.
Docks, dredging and navigational aids	Docks maintain water levels for ship handling (wet docks) or provide dry areas for repair (dry docks).
Tunneling methods and lining	Tunneling methods depend on the ground conditions.
Tunnel services and safety	Safe tunnel operation requires continuous ventilation (to remove exhaust gases), adequate lighting, effective drainage and fire-safety systems.
Airport planning and runway design	Airport planning balances traffic needs with site constraints.
Visual aids and airport safety	Visual aids include runway markings (threshold, centre line) and lighting (approach, runway edge, PAPI) to guide pilots during landing and takeoff.

LEARNING RESOURCES

References

Recommended harbour-tunnel-airport references

1. S. P. Bindra, A Course in Bridge, Tunnel and Railway Engineering.
2. R. Srinivasan, Harbour, Dock and Tunnel Engineering.
3. S. K. Khanna, C. E. G. Justo and A. Veeraragavan, Highway Engineering (Airport section).
4. Current IRC, Railway and Civil Aviation (DGCA/ICAO) manuals.

Approved public harbour-tunnel-airport sources

- https://nptel.ac.in/courses/114106025
- https://nptel.ac.in/courses/105107123

These sources provide the verified study basis while official Course 6013C detail is unavailable; they do not replace official maritime planning data, authorised aviation regulations, certified tunnel engineering or authorised port safety codes.